Part 1: Normalization

Task 1: Normal Form Identification and Violations

The unnormalized table Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price) violates 1NF, 2NF, and 3NF:

1. **1NF Violation:**
    - **Reason:** The seat_numbers column stores multi-valued (non-atomic) data (e.g., "12A, 12B"). 1NF requires every column to contain atomic (indivisible) values.
2. **2NF Violation:**
    - **Reason:** Partial functional dependencies exist. For example, passenger_full_name depends only on part of a composite logical key (passenger_passport_number), rather than the entire booking/seat combination.
3. **3NF Violation:**
    - **Reason:** Transitive dependencies exist among non-key attributes:
        - flight_number $\rightarrow$ airline_name
        - departure_airport_name $\rightarrow$ departure_city
        - arrival_airport_name $\rightarrow$ arrival_city

Task 2: Step-by-Step Normalization

Step 1: 1st Normal Form (1NF)

Eliminate non-atomic values by splitting comma-separated seats into individual rows.

- **Booking_receipt_1NF** (booking_id, passenger_passport_number, seat_number, passenger_full_name, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)
    - **Primary Key:** (booking_id, passenger_passport_number, seat_number)

Step 2: 2nd Normal Form (2NF)

Remove partial dependencies by isolating passenger details into a separate table.

1. **Passenger** (passport_number, passenger_full_name)
    - **Primary Key:** passport_number
2. **Booking_receipt_2NF** (booking_id, seat_number, *passport_number*, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)
    - **Primary Key:** (booking_id, seat_number)
    - **Foreign Key:** passport_number $\rightarrow$ Passenger(passport_number)

Step 3: 3rd Normal Form (3NF)

Remove transitive dependencies by isolating Airline, Airport, Flight, and Booking entities.

1. **Passenger** (passport_number, passenger_full_name)
    - **Primary Key:** passport_number
2. **Airline** (airline_id, airline_name)
    - **Primary Key:** airline_id
3. **Airport** (airport_code, airport_name, city)
    - **Primary Key:** airport_code
4. **Flight** (flight_number, *departure_airport_code*, *arrival_airport_code*, *airline_id*)
    - **Primary Key:** flight_number
    - **Foreign Keys:**
        - departure_airport_code $\rightarrow$ Airport(airport_code)
        - arrival_airport_code $\rightarrow$ Airport(airport_code)
        - airline_id $\rightarrow$ Airline(airline_id)
5. **Booking** (booking_id, *flight_number*, ticket_price)
    - **Primary Key:** booking_id
    - **Foreign Key:** flight_number $\rightarrow$ Flight(flight_number)
6. **Booking_Passenger_Seat** (booking_id, seat_number, *passport_number*)
    - **Primary Key:** (booking_id, seat_number)
    - **Foreign Keys:**
        - booking_id $\rightarrow$ Booking(booking_id)
        - passport_number $\rightarrow$ Passenger(passport_number)

Task 3: Functional Dependencies (FDs)

The following Functional Dependencies justify each step:

- **1NF $\rightarrow$ 2NF (Removing partial dependency):**
    - passenger_passport_number $\rightarrow$ passenger_full_name
- **2NF $\rightarrow$ 3NF (Removing transitive dependencies):**
    - flight_number $\rightarrow$ airline_id, departure_airport_code, arrival_airport_code
    - departure_airport_code $\rightarrow$ departure_airport_name, departure_city
    - arrival_airport_code $\rightarrow$ arrival_airport_name, arrival_city
    - airline_id $\rightarrow$ airline_name
    - booking_id $\rightarrow$ flight_number, ticket_price

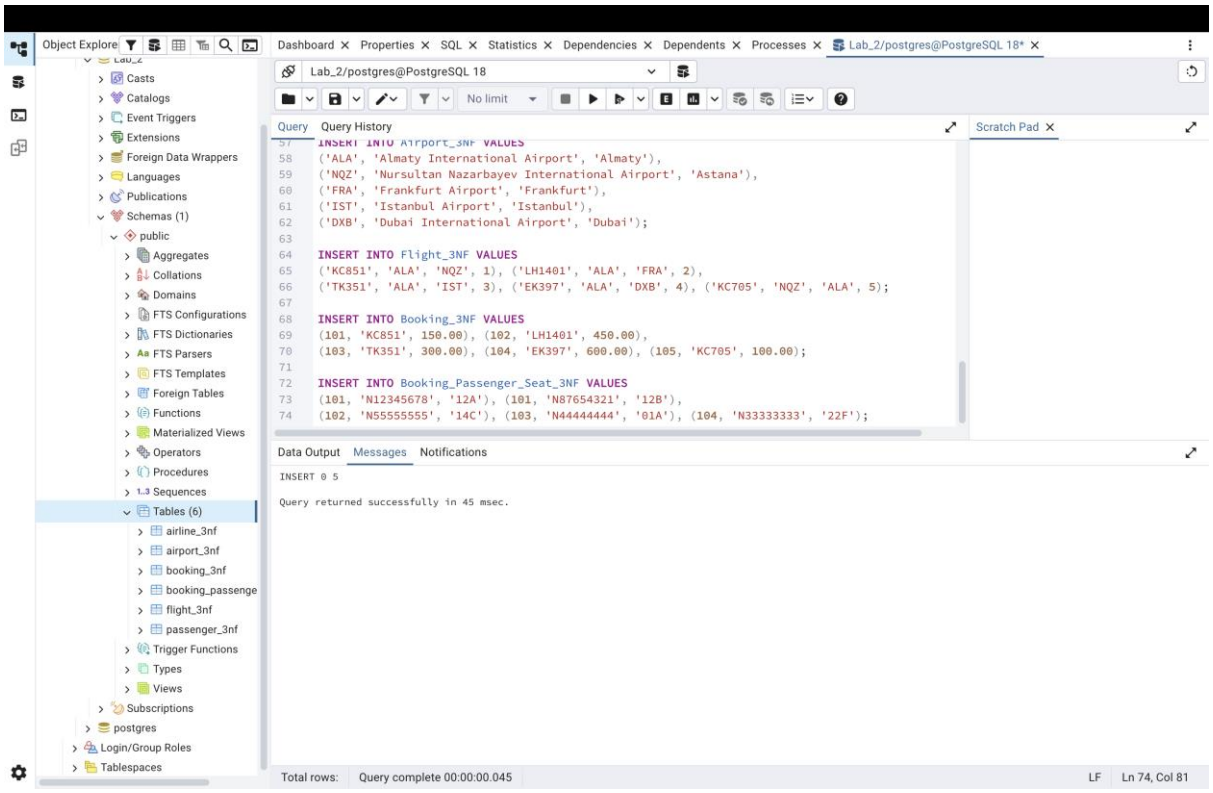
Task 4: 3NF Evaluation of the Flights Table

- **Current Primary Key:** flight_id
- **Functional Dependency:**
    - flight_id $\rightarrow$ sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at
- **Conclusion:**

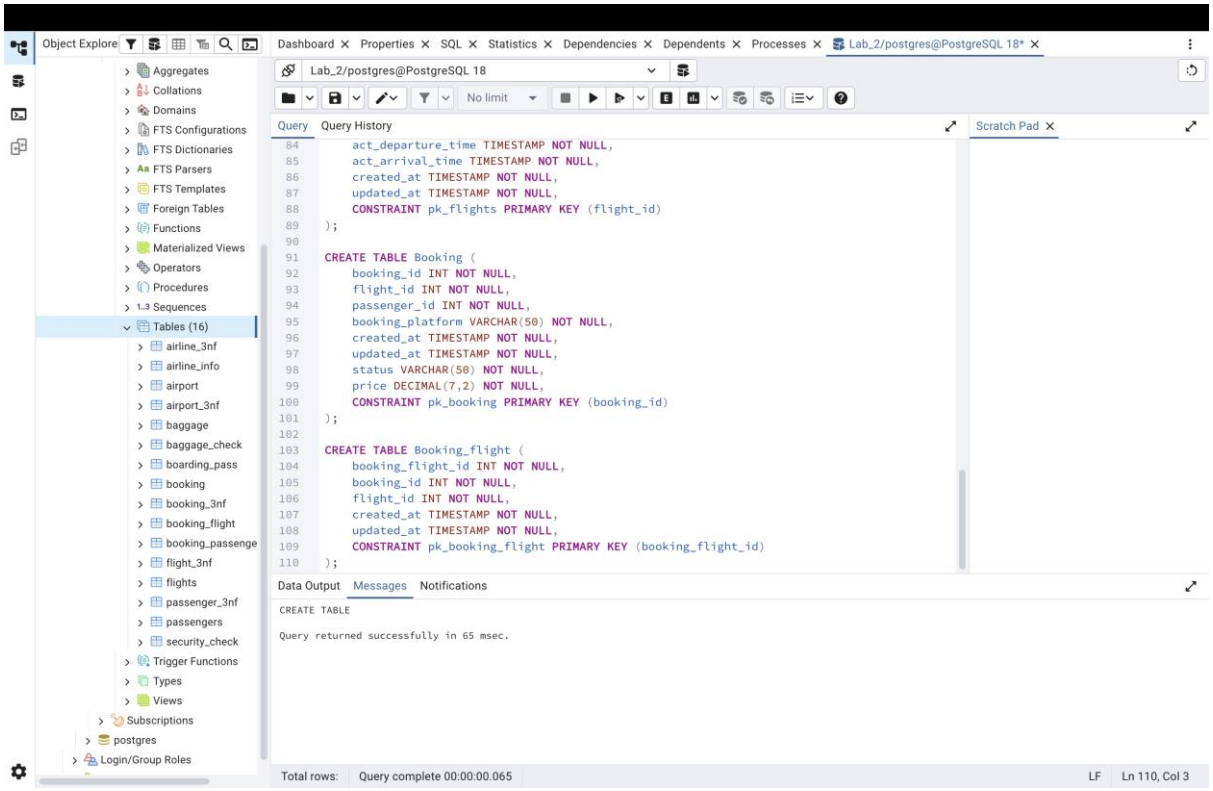
  The Flights table **IS in 3NF**. All non-key attributes directly depend solely on the primary key (flight_id). Foreign keys (departing_airport_id, arriving_airport_id,

airline_id) link to separate referenced tables where detailed attributes (such as airport name or city) are stored. Thus, no transitive dependencies exist between non-key attributes in Flights.

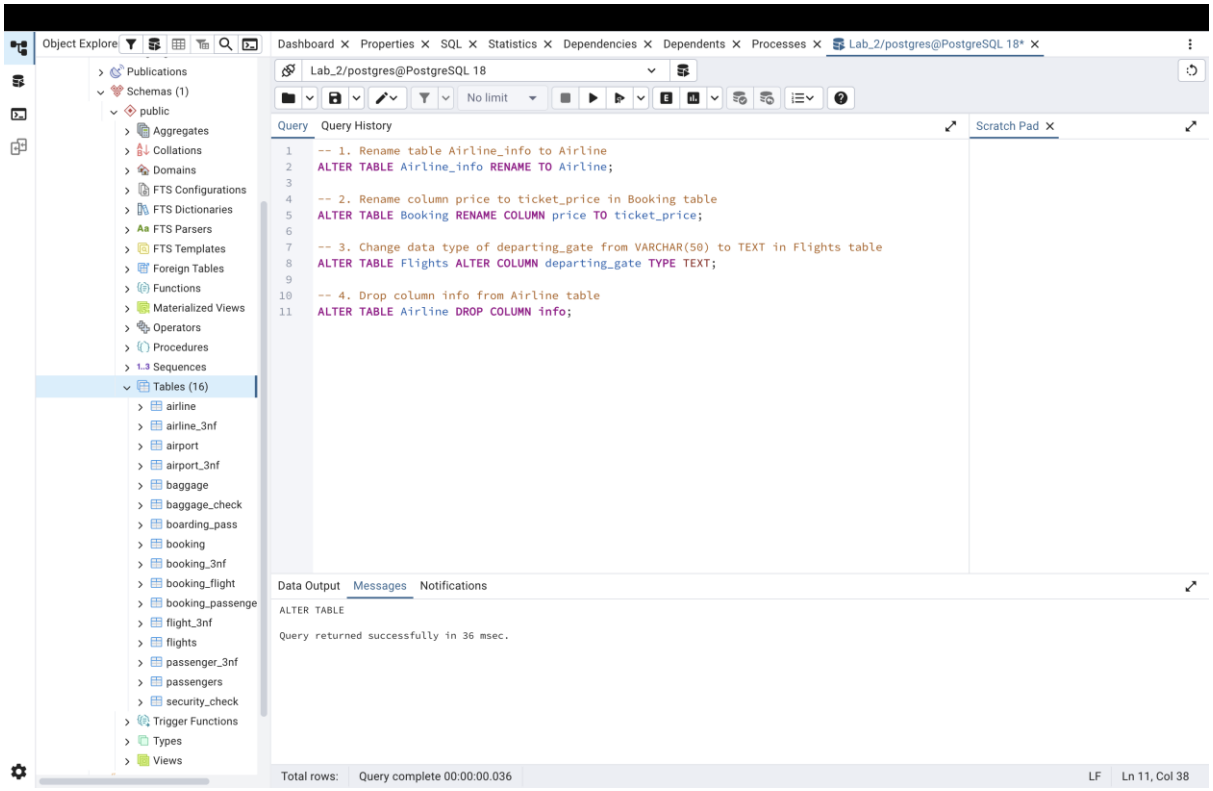
Step 5:



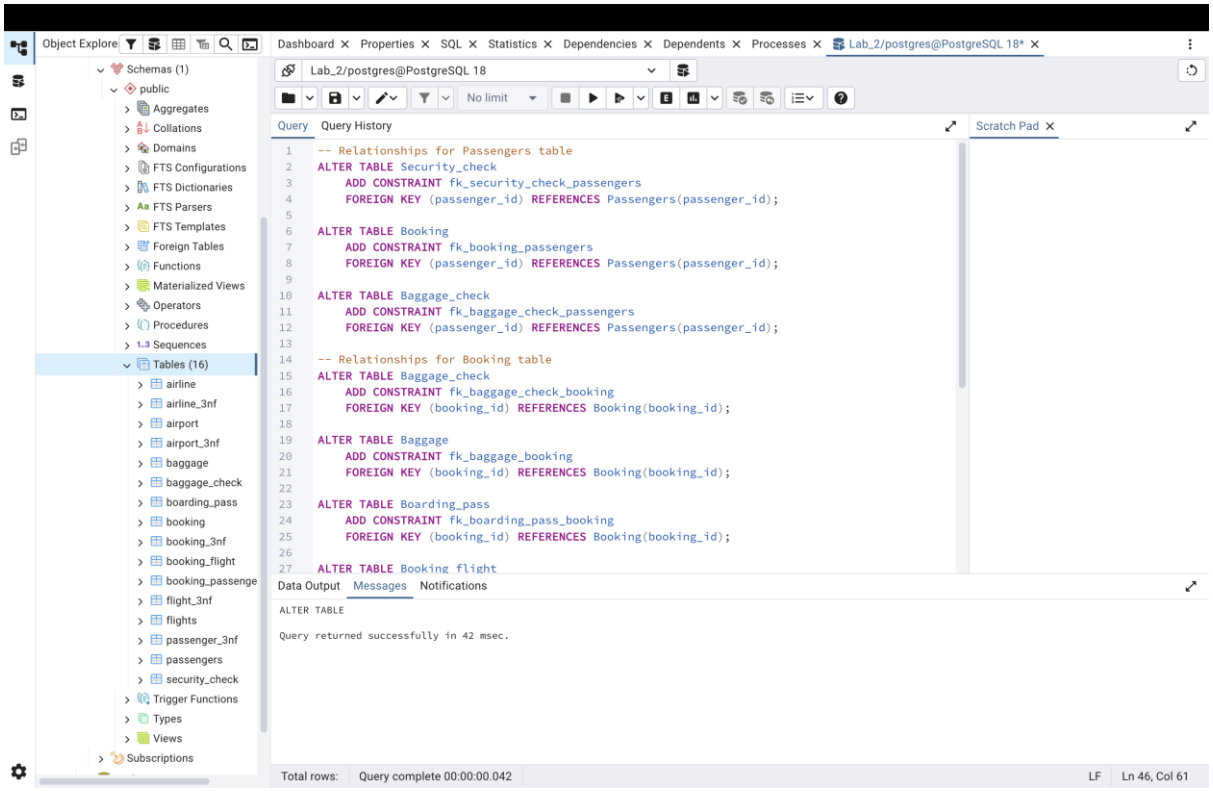
Step 6:

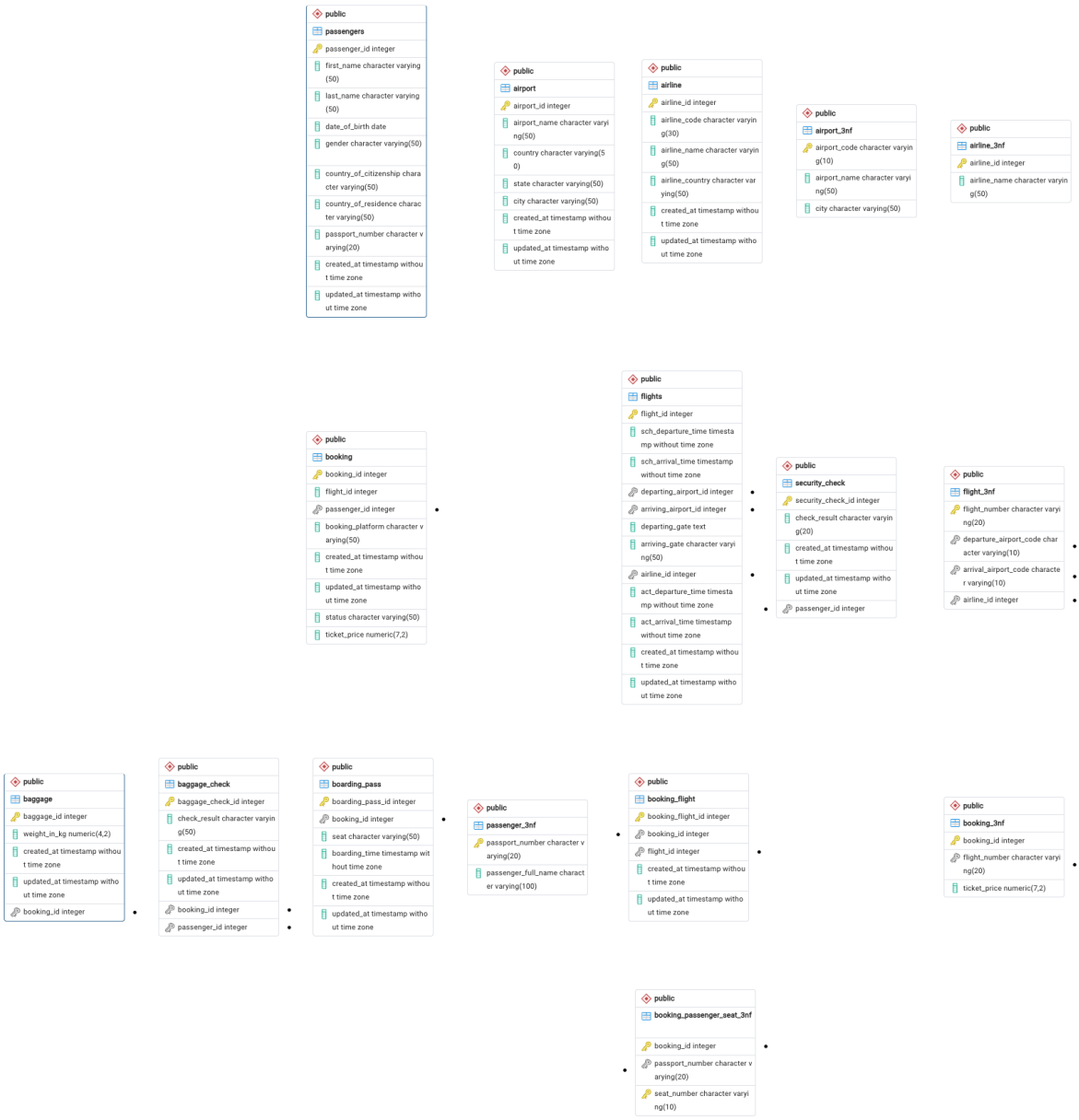

Object Explorer
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Lab_2/postgres@PostgreSQL 18*
Lab_2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline_3nf
airline_info
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenge
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
Subscriptions
postgres
Login/Group Roles
act_departure_time TIMESTAMP NOT NULL,
act_arrival_time TIMESTAMP NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
CONSTRAINT pk_flights PRIMARY KEY (flight_id)
);
CREATE TABLE Booking (
booking_id INT NOT NULL,
flight_id INT NOT NULL,
passenger_id INT NOT NULL,
booking_platform VARCHAR(50) NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
status VARCHAR(50) NOT NULL,
price DECIMAL(7,2) NOT NULL,
CONSTRAINT pk_booking PRIMARY KEY (booking_id)
);
CREATE TABLE Booking_flight (
booking_flight_id INT NOT NULL,
booking_id INT NOT NULL,
flight_id INT NOT NULL,
created_at TIMESTAMP NOT NULL,
updated_at TIMESTAMP NOT NULL,
CONSTRAINT pk_booking_flight PRIMARY KEY (booking_flight_id)
);
Data Output
Messages
Notifications
CREATE TABLE
Query returned successfully in 65 msec.
Total rows:
Query complete 00:00:00.065
LF
Ln 110, Col 3

Step 7:


Object Explorer
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Lab_2/postgres@PostgreSQL 18*
Lab_2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
Publications
Schemas (1)
public
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Collations
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FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
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Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenge
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
Views
-- 1. Rename table Airline_info to Airline
ALTER TABLE Airline_info RENAME TO Airline;
-- 2. Rename column price to ticket_price in Booking table
ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
-- 3. Change data type of departing_gate from VARCHAR(50) to TEXT in Flights table
ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
-- 4. Drop column info from Airline table
ALTER TABLE Airline DROP COLUMN info;
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 36 msec.
Total rows:
Query complete 00:00:00.036
LF
Ln 11, Col 38

Step 8:


Object Explorer
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_3nf
booking_flight
booking_passenge
flight_3nf
flights
passenger_3nf
passengers
security_check
Trigger Functions
Types
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Dependencies
Dependents
Processes
Lab_2/postgres@PostgreSQL 18*
Lab_2/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
-- Relationships for Passengers table
ALTER TABLE Security_check
ADD CONSTRAINT fk_security_check_passengers
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
ALTER TABLE Booking
ADD CONSTRAINT fk_booking_passengers
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
ALTER TABLE Baggage_check
ADD CONSTRAINT fk_baggage_check_passengers
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
-- Relationships for Booking table
ALTER TABLE Baggage_check
ADD CONSTRAINT fk_baggage_check_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
ALTER TABLE Baggage
ADD CONSTRAINT fk_baggage_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
ALTER TABLE Boarding_pass
ADD CONSTRAINT fk_boarding_pass_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
ALTER TABLE Booking flight
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 42 msec.
Total rows:
Query complete 00:00:00.042
LF
Ln 46, Col 61


public
passengers
passenger_id integer
first_name character varying (50)
last_name character varying (50)
date_of_birth date
gender character varying(50)
country_of_citizenship character varying(50)
country_of_residence character varying(50)
passport_number character varying(20)
created_at timestamp without time zone
updated_at timestamp without time zone
public
airport
airport_id integer
airport_name character varying(50)
country character varying(50)
state character varying(50)
city character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone
public
airline
airline_id integer
airline_code character varying(30)
airline_name character varying(50)
airline_country character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone
public
airport_3nf
airport_code character varying(10)
airport_name character varying(50)
city character varying(50)
public
airline_3nf
airline_id integer
airline_name character varying(50)
public
booking
booking_id integer
flight_id integer
passenger_id integer
booking_platform character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone
status character varying(50)
ticket_price numeric(7,2)
public
flights
flight_id integer
sch_departure_time timestamp without time zone
sch_arrival_time timestamp without time zone
departing_airport_id integer
arriving_airport_id integer
departing_gate text
arriving_gate character varying(50)
airline_id integer
act_departure_time timestamp without time zone
act_arrival_time timestamp without time zone
created_at timestamp without time zone
updated_at timestamp without time zone
public
security_check
security_check_id integer
check_result character varying(20)
created_at timestamp without time zone
updated_at timestamp without time zone
passenger_id integer
public
flight_3nf
flight_number character varying(20)
departure_airport_code character varying(10)
arrival_airport_code character varying(10)
airline_id integer
public
baggage
baggage_id integer
weight_in_kg numeric(4,2)
created_at timestamp without time zone
updated_at timestamp without time zone
booking_id integer
public
baggage_check
baggage_check_id integer
check_result character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone
booking_id integer
passenger_id integer
public
boarding_pass
boarding_pass_id integer
booking_id integer
seat character varying(50)
boarding_time timestamp without time zone
created_at timestamp without time zone
updated_at timestamp without time zone
public
passenger_3nf
passport_number character varying(20)
passenger_full_name character varying(100)
public
booking_flight
booking_flight_id integer
booking_id integer
flight_id integer
created_at timestamp without time zone
updated_at timestamp without time zone
public
booking_3nf
booking_id integer
flight_number character varying(20)
ticket_price numeric(7,2)
public
booking_passenger_seat_3nf
booking_id integer
passport_number character varying(20)
seat_number character varying(10)